# Residual Alignment and Conformal-End Mass Certificates for Maximal Asymptotically Hyperbolic Penrose Data

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**Abstract**

We study the deformation interfaces in the assigned manuscript *The Penrose inequality for maximal asymptotically hyperbolic initial data*. Its global existence, transmission, enclosure, and flow machinery is not certified here. For actual smooth graph fields, we derive a coupled residual budget that preserves the source's positive cosmological square and separates the defects relevant to scalar curvature, current flux, and boundary mean curvature. We also retain the differentiated height defect when the height equation is not exact. On an exact conformal hyperbolic end, a static Green identity yields explicit finite-radius error bounds for all four metric mass charges, including nonlinear and discarded-tail costs. A moving-annulus example shows why small subcritical weighted metric error alone does not control mass. These are conditional local certificates, not an existence algorithm or an independent proof of the source's global inequality. Contribution assessment: C2/5, modest source-specific residual and mass-error refinements of classical identities, with the global geometric construction and its uniform estimates still imported.

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# 1 The geometric problem and its proof boundary

## 1.1 Exact conventions

The assigned source [1] considers a smooth connected orientable three-manifold $\Omega$, complete with its nonempty compact smooth boundary $S$, with one spherical end. In that end,

$$b = \frac{\mathrm{d}r^2}{1+r^2} + r^2\sigma, \qquad V_0 = \sqrt{1+r^2}, \qquad V_i = r\omega_i \quad (1 \le i \le 3). \tag{1}$$

Here $\sigma$ is the unit round metric and $\omega \in S^2$. We use $\Delta = \operatorname{div}\nabla$, so its integration-by-parts quadratic form is nonpositive. The boundary normal $\nu$ points into $\Omega$ toward its end; the outward normal of a truncated exterior at $S$ is therefore $-\nu$. Mean curvature is the sum, not the average, of principal curvatures.

For $0 < \alpha < 1$ and $3/2 < \tau < 3$, the source requires $g - b \in C^{2,\alpha}_\tau$ and $K \in C^{1,\alpha}_\tau$ in physical $b$-tensor norms and uniformly controlled background charts. These are not componentwise Euclidean decay statements. With cosmological constant $-3$ and the source's convention for $K$, maximality and the constraints are

$$\operatorname{tr}_g K = 0, \quad 16\pi\mu = R_g + 6 - |K|_g^2, \quad 8\pi J_i = \nabla^j K_{ij}, \quad \mu \ge |J|_g. \tag{2}$$

Weighted integrability of $V_0(\mu + |J| + |K|^2)$ on the end is an additional hypothesis. Only the outward expansion is assumed nonpositive:

$$\theta_+ = H + \operatorname{tr}_S K \le 0.$$

Since $\operatorname{tr}_S K = -K(\nu,\nu)$, this says $H \le K(\nu,\nu)$; it does not say that $S$ is minimal or that the inward expansion has a specified sign.

For $e = g - b$ define the linear mass current

$$\mathbb{U}(V,e) = V(\operatorname{div}_b e - \mathrm{d}\operatorname{tr}_b e) + (\operatorname{tr}_b e)\,\mathrm{d}V - e(\nabla_b V, \cdot), \qquad p_\mu = \frac{1}{16\pi}\lim_{R\to\infty}\int_{r=R}\mathbb{U}(V_\mu, e)(\nu_b)\,\mathrm{d}A_b. \tag{3}$$

The four charges are metric AH charges, not the ADM momentum of an asymptotically flat initial-data set. The original charge covector is assumed future timelike, and $m_{\mathrm{AH}} = \sqrt{p_0^2 - |\mathbf{p}|^2}$.

The minimum area $A_*$ is the infimum of the *entire intrinsic boundary area* of connected closed end-side domains whose interiors lie in the interior of $\Omega$. The domain $\Omega$ itself is allowed and contributes $S$. One must not replace this boundary with its relative frontier in $\Omega$, which could be

empty. Disconnected boundary components, arbitrary genus, and compact topology are allowed. Writing $a = \sqrt{A_*/(4\pi)}$, the source claims

$$m_{\mathrm{AH}} \geq \frac{a + a^3}{2}. \tag{4}$$

This report does not independently establish that global statement.

### 1.2 What the source actually needs

The architecture has two different deformations. First, maximal data are reduced to a time-symmetric scalar-curvature problem using an AH graph, conformal factor, boundary selector, and enclosing minimal surface. Second, time-symmetric AH data are compared to an AF exterior using a nonlinear electrostatic stress, an attachment, a compact graph transmission problem, smoothing, and two different minimizing enclosures. The synthetic tensor in that second construction is not the original physical $K$.

| Source module | Role and retained obligation |
|---|---|
| Preliminaries; graph calculus | Mass covariance, actual boundary domains, static graph and conformal identities, gradient and linearization estimates. |
| Area transfer | Weak flow, topology-dependent curvature bound, projection and weighted bad-volume criterion. |
| Maximal reduction | Positive finite-parameter solutions, end estimates, contact selector, free enclosure and weighted volume control. |
| Stress extension | Conformal-end approximation; nonlinear electrostatic solution, charge saturation, and scalar-flat AF attachment. |
| Compact construction; transmission | Fixed-parameter estimates, full boundary and interface operators, continuation, then outer exhaustion. |
| Scalar errors; completion | Unsafe-set exclusion, corner smoothing, mass budget, two enclosures and the AF Penrose theorem. |

The source's time-symmetric comparison gives a lower bound for the *time component in a fixed chart*, without requiring the comparison covectors to be timelike. One first chooses a rest chart for the original timelike covector and then keeps that chart fixed. Rest-frame invariance is not a license to boost each approximate metric separately.

The final finite-stage chain, for $A_* > 0$, is

$$p_0(g) \geq \frac{a^3}{2} + \frac{a}{2}\sqrt{1 - \varepsilon_A} - 2\varepsilon_m. \tag{5}$$

It follows from the source's attachment bound $m_o \leq p_0(g) - a^3/2 + \varepsilon_m$, its AF comparison $m_{\mathrm{AF}} \leq m_o + \varepsilon_m$, and an enclosing area at least $(1 - \varepsilon_A)4\pi a^2$. The mass theorem is applied outside the enclosure that also contains the unsafe set; area transfer uses the cap-only enclosure. Their ordered perimeters connect the two. The source's temporary conformal change for starting the flow preserves initial area and is not substituted into the mass theorem.

The choices are sequential: preliminary approximation and attachment, then finite internal parameters and strengths, then exterior exhaustion at those parameters, with separately small smoothing. Arbitrarily accurate finite comparisons suffice for (5); a limit of all comparison metrics is neither asserted nor needed. Finite matrices or locally invertible frozen operators do not supply any of these global choices.

## 2 Actual graph fields and aligned defects

### 2.1 Constitutive geometry

Work first on a smooth compact truncated exterior $U$ with inner boundary $S$ and outer boundary $S_R$. All fields in this section are smooth, $g$ is positive definite, $K$ is symmetric and trace free, and $N, w > 0$. Define $s > 0$ by

$$\begin{aligned} s^2 + s^4 |\,\mathrm{d}f|_g^2 = N^2, \quad p = s^2 \nabla_g f, \quad v = p/N, \\ \bar{g} = g + s^2\,\mathrm{d}f \otimes \mathrm{d}f, \quad x = sw, \quad h = x\bar{g}, \quad A = Ns\bar{g}^{-1}. \end{aligned} \tag{6}$$

Thus $|v|_g < 1$, $\bar{g}^{-1} = g^{-1} - v \otimes v$, $\mathrm{d}V_{\bar{g}} = (N/s)\,\mathrm{d}V_g$. Tensors $A\,\mathrm{d}w$ and $Kp$ mean vectors with the indicated metric contractions. Put

$$\pi = N^{-1} \operatorname{sym} \nabla p^\flat, \quad \ell = N^{-1} \operatorname{div}_g p, \quad \lambda = \pi - K, \quad D = R_g + 6 - |K|_g^2 - 2(\operatorname{div}_g K) \cdot v.$$

The symmetrization includes the factor $1/2$. Then $\operatorname{tr}_g \pi = \ell$; under (2), $D = 16\pi(\mu - J \cdot v) \geq 0$.

For scalar fields $\Sigma, P$, introduce the *actual* residuals

$$\begin{aligned} B_0 &= \nabla N - Kp, \qquad Q_0 = B_0 + A\,\mathrm{d}w, \\ e_N &= \operatorname{div}_g B_0 - 3N - \Sigma, \\ e_w &= -\operatorname{div}_g(A\,\mathrm{d}w) - 3N - \Sigma x - P. \end{aligned} \tag{7}$$

There is no assumption here that a finite list of residual moments determines these functions. The residual must include all coefficient variation, including the dependence of $s, A$ on $N$, $\mathrm{d}f$.

**Theorem 2.1** (Coupled scalar, current, and boundary budget)**.** *With* (6)–(7)*, set*

$$Z = x(D + |\lambda|_g^2) + \frac{3}{2x} |\,\mathrm{d}x|_{\bar{g}}^2 + 6(x-1)^2 + \frac{2P}{N}.$$

*The following identities hold:*

$$x^2(R_h + 6) = Z + \frac{2}{N}(e_w - x e_N) - x\ell^2 + \frac{2x}{N} \operatorname{div}_g(\ell p), \tag{8}$$

$$\operatorname{div}_g Q_0 = -(x-1)\Sigma - P + e_N - e_w. \tag{9}$$

*In particular, when* $\operatorname{div}_g p = 0$*, the height-defect terms in* (8) *vanish. On an inner constant-height boundary, assume* $v = -z\nu$*,* $z \geq 0$*, and write* $k = K(\nu, \nu)$*,*

$$B_0 \cdot \nu = -b_S + \beta_B, \qquad (A\,\mathrm{d}w) \cdot \nu = \beta_w.$$

*Then, without requiring* $\ell = 0$*,*

$$s\sqrt{x} H_h = N(H - zk) - b_S + \beta_B + \beta_w/x + \ell N z. \tag{10}$$

*On a domain whose boundaries are precisely* $S, S_R$*,*

$$-\int_{S_R} Q_0 \cdot \nu_R \,\mathrm{d}A_g = \int_S (b_S - \beta_B - \beta_w)\,\mathrm{d}A_g + \int_U \left[(x-1)\Sigma + P + e_w - e_N\right] \mathrm{d}V_g. \tag{11}$$

*Proof.* We retain the source's geometric identities, deriving their residual specialization rather than postulating a new scalar equation. For general trace $\ell$, set $B = B_0 - \ell p$, $C = \lambda - \ell g$, $\Phi = \operatorname{div}_g B$, and $\Psi = -\operatorname{div}_g(A\,\mathrm{d}w)$. The graph calculation in [1, graph calculus] gives

$$R_{\bar{g}} = D - 6 + |\lambda|_g^2 - \ell^2 - \frac{2}{N}\operatorname{div}_g(Cp), \tag{12}$$

$$N\Delta_{\bar{g}}x + x\operatorname{div}_g(Cp) = x\Phi - \Psi. \tag{13}$$

For completeness, the second cancellation can be checked directly: $\bar{\nabla}\log s = \nabla\log N - \pi(v)$ implies $A\,\mathrm{d}w = N\bar{\nabla}x - x\nabla N + x\pi p$. Hence $B + A\,\mathrm{d}w = N\bar{\nabla}x + xCp + (1-x)B$. Its divergence is both $\Phi - \Psi$ and $N\Delta_{\bar{g}}x + x\operatorname{div}_g(Cp) + (1-x)\Phi$; all first derivatives of $x$ cancel. The first identity is the source's static-graph Gauss–Codazzi formula, completed with the square $|\pi - K|^2$; it does not assume the residual equations vanish.

The three-dimensional conformal formula is

$$xR_h = R_{\bar{g}} - 2x^{-1}\Delta_{\bar{g}}x + \frac{3}{2}x^{-2}|\,\mathrm{d}x|_{\bar{g}}^2.$$

Substitute (12)–(13). The resulting right side is $D-6+|\lambda|^2-\ell^2-2\Phi/N+2\Psi/(Nx)+(3/2)|\,\mathrm{d}\log x|_{\bar{g}}^2$. Now

$$\Phi = 3N + \Sigma + e_N - \operatorname{div}_g(\ell p), \qquad \Psi = 3N + \Sigma x + P + e_w.$$

Multiply by $x$ and add $6x^2$. The cosmological terms are $6x^2 - 12x + 6 = 6(x-1)^2$, proving (8). The divergence of the defining expression for $Q_0$ proves (9).

The source's constant-height identity is $s\sqrt{x}H_h = N(H - k_T\widehat{z}) + B\cdot\nu + (A\,\mathrm{d}w)\cdot\nu/x$, with $k_T = \operatorname{tr}_S K = -k$, $\widehat{z} = -z$. Since $B\cdot\nu = B_0\cdot\nu + \ell N z$, this is (10). Finally, the inner contribution to the outward flux of $Q_0$ is $b_S - \beta_B - \beta_w$. Integrate (9) with that sign and rearrange to obtain (11). □

## 2.2 One-sided acceptance conditions

Suppose DEC holds, $P \geq 0$, and the height equation is exact. Then $Z \geq 0$ and a sufficient, explicitly pointwise condition for $R_h \geq -6$ is

$$(e_w - xe_N)_- \leq NZ/2. \tag{14}$$

This uses the actual positive square and matter margin. It does not discard them and require both equations to be solved exactly. If a lower bound $Z \geq Z_*$ is certified, one can replace the right side by $NZ_*/2$, provided $Z_* \geq 0$. For $H_h \leq 0$ at the boundary it suffices that $\beta_B + \beta_w/x \leq b_S - N(H - zk)$. For strict negativity, use a strict positive remaining margin. Neither DEC nor $H \leq k$ by itself gives that last margin: $z < 1$, and the source's selector is still a real input.

For a nonzero height residual, the scalar quantity which must instead be bounded from below is

$$-x\ell^2 + \frac{2x}{N}\operatorname{div}_g(\ell p) = x\ell^2 + \frac{2x}{N}p\cdot\nabla\ell. \tag{15}$$

The equality uses $\operatorname{div}_g p = N\ell$. Thus a small $C^0$ or $L^2$ height residual alone supplies no bound on this expression. For example $\ell_j(t) = j^{-1}\sin(j^2 t)$ tends uniformly to zero but $\partial_t\ell_j$ has supremum $j$; this is a diagnostic for the missing derivative estimate, not a constructed solution of the graph system. The boundary term $\ell N z$ must likewise be retained. This is why enforcing the divergence equation in a genuinely conservative reconstruction can be valuable.

**Corollary 2.2** (Aligned residuals)**.** *Suppose $\ell = 0$. If $e_w = xe_N + \rho$, then the scalar error is $2\rho/N$, while the bulk current error is $(x-1)e_N + \rho$. In particular, if $|x-1| \le \delta$ and all integrals below are finite,*

$$\left| \int_U (e_w - e_N)\, \mathrm{d}V_g \right| \le \delta \|e_N\|_{L^1(U,g)} + \|\rho\|_{L^1(U,g)}. \tag{16}$$

*The finite current budget also includes $\|\beta_B + \beta_w\|_{L^1(S,g)}$.*

*Proof.* Substitution and the integral triangle inequality prove the assertions. Importantly, the boundary scalar condition uses $\beta_B + \beta_w/x$, whereas the flux condition uses $\beta_B + \beta_w$. □

If $e_w = xe_N$ and $\Sigma + e_N \ge 0$, this is exactly a change of the penalty $\Sigma$, an observation already implicit in the source's cancellation. The useful error statement is the quantitative extension to signed residuals, nonzero $\rho$, and independently bounded boundary and end contributions. It is not a new positivity mechanism.

As a rational finite budget example, let $N = 1$, $x = 1 + \delta$ on a unit-volume test region, $e_N = L$, $e_w = xL$, and zero boundary defects. The scalar residual is zero and the absolute flux error is $|\delta L|$, while independent absolute equation bounds give $(2 + \delta)|L|$ when $\delta \ge 0$. For $\delta = 1/100$, $L = 1$, these are $1/100$ and $201/100$. These scalar assignments demonstrate the ledger's cancellation; they are not asserted to arise from a globally admissible geometric solution.

### 2.3 Where the mass conversion enters

The current identity alone is not a mass formula. The source separately proves end decay for $N, f, w$, weighted scalar integrability, and

$$\lim_{R\to\infty} \int_{S_R} Q_0 \cdot \nu_R \, \mathrm{d}A_g = -8\pi\{p_0(h) - p_0(g)\}$$

for its exact-height AH construction. An approximate solution may use this conversion only if its own corresponding end estimates or a quantified difference from such a flux are certified. Under this explicit additional premise, integrability of all defect terms, and convergence of the boundary and source terms, (11) gives the exact mass increment with its $e_w - e_N$ and boundary corrections, divided by $8\pi$. It does not control the existence of a free enclosing surface, weighted bad volume, or source penalty strengths.

## 3 Conformal-end mass as a static residual moment

### 3.1 Exact end identity

The source's preliminary time-symmetric approximation has an exact conformal end $g_u = (1+u)^4 b$. This is a separate variable and metric from the graph factor $x$ of Section 2. On $E_R = \{r \ge R\}$, $R \ge 1$, assume $u$ is smooth, $1 + u > 0$, and, uniformly in angle with two physical derivatives,

$$u(r,\omega) = \alpha(\omega) r^{-3} + O(r^{-3-\xi}), \qquad \xi > 0. \tag{17}$$

All assertions are in the one fixed chart (1). Set $L = -\Delta_b + 3$ and assume $V_0 Lu \in L^1(E_R, \mathrm{d}V_b)$. Define the signed finite-radius flux

$$F_\mu(R; u) = \int_{r=R} (V_\mu \partial_{\nu_b} u - u \partial_{\nu_b} V_\mu)\, \mathrm{d}A_b, \qquad \omega_0 = 1.$$

**Proposition 3.1** (Mass and static Green identity)**.** *Under these hypotheses, all four charges exist and*

$$p_\mu(g_u) = \frac{2}{\pi}\int_{S^2} \alpha(\omega)\omega_\mu \,\mathrm{d}A_\sigma, \tag{18}$$

$$p_\mu(g_u) = \frac{1}{2\pi}\left[\int_{E_R} V_\mu Lu \,\mathrm{d}V_b - F_\mu(R;u)\right]. \tag{19}$$

*Proof.* If $e = \psi b$ in dimension three, (3) reduces to $\mathbb{U}(V,e) = -2V\,\mathrm{d}\psi + 2\psi\,\mathrm{d}V$. For $\psi = (1+u)^4 - 1 = 4u + O(u^2)$, the contribution of $\psi - 4u$ to the sphere integral tends to zero: its physical size is $O(r^{-6})$, the static potential and its gradient are $O(r)$, and sphere area is $4\pi r^2$. Thus the mass equals $-(2\pi)^{-1}\lim F_\mu$. Because $\nu_b = \sqrt{1+r^2}\,\partial_r$, direct substitution of (17) gives $\lim F_\mu = -4\int \alpha\omega_\mu \,\mathrm{d}A_\sigma$.

Each $V_\mu$ satisfies $\Delta_b V_\mu = 3V_\mu$. The divergence of $V_\mu \nabla_b u - u\nabla_b V_\mu$ is $-V_\mu Lu$. Integrating on $R < r < T$ gives $F_\mu(T;u) - F_\mu(R;u) = -\int_{R<r<T} V_\mu Lu$. Take $T \to \infty$, using $|V_i| \le V_0$ and the integrability hypothesis. This proves both formulas with the stated orientation. □

For $u = cr^{-3}$ on the end, the normalization is $p_0 = 8c$, $\mathbf{p} = 0$, not $2c$ or $c$. In fact $L(cr^{-3}) = -6cr^{-5}$ and

$$F_0(R; cr^{-3}) = -4\pi c(4 + 3R^{-2}), \qquad \int_{E_R} V_0 Lu \,\mathrm{d}V_b = -12\pi c R^{-2}.$$

Their difference is $16\pi c$ at every $R$, verifying (19) without taking a finite-radius flux to be the mass by itself.

### 3.2 A nonlinear mass-error certificate

The source's conformal equation becomes, on its exact $b$ end,

$$8Lu + \mathcal{N}(u) = S, \qquad \mathcal{N}(u) = 60u^2 + 60u^3 + 30u^4 + 6u^5. \tag{20}$$

Indeed $\mathcal{N}(u) = 6[(1+u)^5 - 1 - 5u]$. The prescribed forcing $S$ is the right side of the conformally rewritten scalar equation; it is not automatically $R_{g_u} + 6$. In the source approximation it equals its positive prescribed scalar density before division by $(1+u)^5$.

**Theorem 3.2** (Explicit all-charge end budget)**.** *Let $u, \widehat{u}$ each satisfy the end and positivity hypotheses of Proposition 3.1. Assume* (20) *for $u$ and define*

$$e = 8L\widehat{u} + \mathcal{N}(\widehat{u}) - S, \qquad d = u - \widehat{u}.$$

*Suppose, on the* entire *end $r \ge R \ge 1$,*

$$|u|, |\widehat{u}| \le Mr^{-j}, \quad |d| \le \varrho r^{-j}, \quad |e| \le e_* r^{-q}, \qquad j > 3/2, \quad q > 3. \tag{21}$$

*Then, for $\mu = 0, 1, 2, 3$,*

$$\begin{aligned} |p_\mu(g_u) - p_\mu(g_{\widehat{u}})| \le{}& \frac{|F_\mu(R;d)|}{2\pi} + \frac{e_*}{4}\frac{R^{3-q}}{q-3} \\ &+ \frac{\varrho}{4}\sum_{k=1}^{4} c_k M^k \frac{R^{3-(k+1)j}}{(k+1)j - 3}, \end{aligned} \tag{22}$$

$$(c_1, c_2, c_3, c_4) = (120, 180, 120, 30).$$

*All denominators are strictly positive. No claim that $\varrho$ follows from a finite linear system is part of the theorem.*

*Proof.* Subtract the two equations: $8Ld = -e - [\mathcal{N}(u) - \mathcal{N}(\widehat{u})]$. Subtract (19) for the two fields. The exact signed formula is

$$\Delta p_\mu = -\frac{F_\mu(R;d)}{2\pi} - \frac{1}{16\pi}\int_{E_R} V_\mu\{e + \mathcal{N}(u) - \mathcal{N}(\widehat{u})\}\,\mathrm{d}V_b. \tag{23}$$

For real arguments with absolute value at most $Mr^{-j}$, the mean value theorem and $\mathcal{N}'(t) = 120t + 180t^2 + 120t^3 + 30t^4$ give

$$|\mathcal{N}(u) - \mathcal{N}(\widehat{u})| \le \varrho \sum_{k=1}^{4} c_k M^k r^{-(k+1)j}.$$

Exactly, not just asymptotically, $V_0\,\mathrm{d}V_b = r^2\,\mathrm{d}r\,\mathrm{d}A_\sigma$. Use $|V_\mu| \le V_0$, integrate $\int_R^\infty r^{2-a}\,\mathrm{d}r = R^{3-a}/(a-3)$ for $a > 3$, and use $\mathrm{Area}(S^2) = 4\pi$. The factor $4\pi/(16\pi) = 1/4$ gives (22). □

The dependence on $j > 3/2$ is structural: the first nonlinear term has decay $r^{-2j}$ and the static mass weight requires $2j > 3$. At the borderline, the absolute majorant has a logarithmic divergence. Actual signed cancellations can do better, but cannot be presumed from a uniform norm bound.

If the sphere values satisfy $|d| \le a_R$, $|\partial_{\nu_b} d| \le b_R$, elementary bounds give

$$\frac{|F_0(R;d)|}{2\pi} \le 2R^2(V_0(R)b_R + Ra_R), \tag{24}$$

$$\frac{|F_i(R;d)|}{2\pi} \le 2R^2(Rb_R + V_0(R)a_R). \tag{25}$$

Here $\partial_{\nu_b} V_0 = R$ and $\partial_{\nu_b} V_i = V_0\omega_i$. Physical normal derivatives, not $\partial_r$ without its factor, are required.

As a concrete rational coefficient budget, take $j = 2$, $q = 4$, $R = 10$, $M = 1/100$, $\varrho = 1/1000$, $e_* = 1/100$. The four-term nonlinear contribution is less than 0.000030002; the residual tail is exactly 0.00025. Thus the nonboundary error is less than 0.000280002. This is a conditional certificate for those specified enclosures, not a computed AH solution. The boundary flux must still be added.

### 3.3 Signed panels and discarded tails

Formula (23) also permits signed integration on $R \le r \le T$ with certified quadrature errors. Let $I_\mu$ be an approximation to the signed integral there, with error at most $\epsilon_\mu$, and let $\widehat{F}_\mu$ approximate $F_\mu(R;d)$ with error at most $\epsilon_{\partial,\mu}$. The charge error is centered at $-\widehat{F}_\mu/(2\pi) - I_\mu/(16\pi)$ with radius

$$\frac{\epsilon_{\partial,\mu}}{2\pi} + \frac{\epsilon_\mu}{16\pi} + \frac{e_*}{4}\frac{T^{3-q}}{q-3} + \frac{\varrho}{4}\sum_{k=1}^{4} c_k M^k \frac{T^{3-(k+1)j}}{(k+1)j-3}.$$

The proof splits the exact integral and applies the preceding tail bound only to $r > T$. This can preserve actual sign cancellation that a whole-end absolute bound loses. It requires the actual nonlinear difference, or a certified enclosure of it, on the panels. Moment reconstruction without a remainder bound cannot provide that enclosure. No numerical speedup is claimed.

## 4 Obtaining a state enclosure, and what does not suffice

### 4.1 The original Ram boundary and inverse requirements

The freshly read original private proofs [6, 7] distinguish a complete PDE operator from an interior differential expression. A correction must lie in a specified Banach domain with the actual boundary

conditions, decay, and any gauge imposed. A Green operator for a free-space problem is not automatically an inverse for a boundary-value problem. Finitely many tested residuals only give a lower bound on a full dual residual norm.

For example, in the source's preliminary conformal problem on $(\Omega, g_B)$, a natural correction domain is a weighted $C^{2,\alpha}_j$ space with homogeneous Neumann trace at $S$, $3/2 < j < 3$, after any nonhomogeneous trace has been lifted. The actual nonlinear interior expression is

$$\mathcal{F}_B(u) = -8\Delta_{g_B} u + (R_{g_B} + 6)(1+u) + 6[(1+u)^5 - (1+u)] - S_B.$$

Its derivative, with that same domain, is

$$D\mathcal{F}_B(u)z = -8\Delta_{g_B} z + [R_{g_B} + 6 + 24 + \mathcal{N}'(u)]z.$$

This retains the zeroth-order geometric term and the nonlinear variation. No gauge is added gratuitously: if another formulation has a kernel, its kernel and compatibility conditions must instead be specified and removed legitimately. The source's positive zeroth-order barrier excludes its relevant homogeneous kernel.

Here is the precise classical local certificate used as a conditional interface, not as a substitute for the source's existence theorem. Let $X, Y$ be Banach spaces, $\mathcal{F} : X \supset \mathcal{O} \to Y$ continuously differentiable, and $B : Y \to X$ a bounded linear *isomorphism*. On the closed ball $\overline{B}_X(\widehat{u}, r) \subset \mathcal{O}$, suppose

$$\|B\mathcal{F}(\widehat{u})\|_X \le \eta, \qquad \sup_{\|v-\widehat{u}\|_X \le r} \|I - BD\mathcal{F}(v)\|_{X\to X} \le q < 1, \qquad \eta + qr \le r. \tag{26}$$

Then $v \mapsto v - B\mathcal{F}(v)$ maps the ball to itself and is $q$-Lipschitz, by integration of its derivative along segments. Banach's fixed-point theorem gives a unique fixed point in the ball; injectivity of $B$ makes it a zero of $\mathcal{F}$. The estimate $\|u - \widehat{u}\|_X \le \eta/(1-q)$ follows by comparing the fixed point to the center. Uniqueness is only within the certified ball.

To use this in Theorem 3.2, the $X$-norm must dominate $\sup r^j|u - \widehat{u}|$; if its embedding constant is $C_X$, take $\varrho = C_X\eta/(1-q)$. Positivity of $1+u$, the end expansion (17), and integrability must also be proved. The ball estimate alone in weight $j < 3$ does not prove the mass expansion or continuity of mass. The source proves that expansion from its end equation and excludes the growing indicial mode. An approximate inverse of a finite matrix does not verify (26) for the infinite boundary operator.

### 4.2 A moving-annulus obstruction

**Proposition 4.1** (Subcritical weighted accuracy does not control mass)**.** *Fix $c > 0$, $0 < \alpha < 1$, and $\tau < 3$. On a hyperbolic exterior $r \ge 1$, let $\chi \in C^\infty([0,\infty))$ equal zero on $[0,1]$ and one on $[2,\infty)$, with $0 \le \chi \le 1$. Define*

$$u_R(r) = c\,\chi(r/R)r^{-3}, \qquad g_R = (1+u_R)^4 b.$$

*Then $g_R = b$ on every prescribed compact set for large $R$, and $\|g_R - b\|_{C^{2,\alpha}_\tau} \to 0$, but $p_0(g_R) = 8c$ and $\mathbf{p}(g_R) = 0$ for every $R$. No scalar-curvature or constraint sign is claimed in the transition annulus.*

*Proof.* On $R \le r \le 2R$, physical radial derivatives are comparable to $r\partial_r$. Each derivative of $\chi(r/R)$ with these physical derivatives is uniformly bounded. Thus $u_R$ and its fixed-order physical derivatives are $O(cR^{-3})$ there. Outside that annulus they are $O(cr^{-3})$, and they vanish inside. Uniform chart estimates, including one extra derivative to control the $\alpha$-seminorm, give $\|g_R - b\|_{C^{2,\alpha}_\tau} = O_c(R^{\tau-3})$. The end coefficient in (17) is identically $c$, so Proposition 3.1 gives the stated charges. □

This explicitly illustrates a warning already made in the source's conformal approximation proof: weighted local convergence below the mass order is insufficient. The source uses weighted scalar integrability and a separate integrated mass expansion to overcome it. Our formula (23) is a quantitative version on the exact conformal end. The moving annulus is not a counterexample to the Penrose claim: it was not constructed to satisfy its energy and horizon hypotheses.

### 4.3 Area and rest-frame errors

When two metrics are conformal to the same $b$ on a domain, and $|u - \widehat{u}| \le \epsilon(1 + \widehat{u})$ there with $0 \le \epsilon < 1$, every two-dimensional area element satisfies

$$(1-\epsilon)^4 \, \mathrm{d}A_{g_{\widehat{u}}} \le \mathrm{d}A_{g_u} \le (1+\epsilon)^4 \, \mathrm{d}A_{g_{\widehat{u}}}.$$

Taking infima over the same admissible end-side cuts preserves these bounds. An enclosure on an end alone does not give this comparison for cuts in an uncontrolled compact region.

If a mass-charge enclosure gives $|p_0 - \widehat{p}_0| \le e_0$ and $|\mathbf{p} - \widehat{\mathbf{p}}| \le e_{\mathrm{sp}}$, with $\widehat{p}_0 - e_0 > |\widehat{\mathbf{p}}| + e_{\mathrm{sp}}$, then

$$m_{\mathrm{AH}}(p) \ge \sqrt{(\widehat{p}_0 - e_0)^2 - (|\widehat{\mathbf{p}}| + e_{\mathrm{sp}})^2}. \tag{27}$$

The proof is monotonicity of the positive square root and the triangle inequality. Componentwise charge errors $e_i$ yield $e_{\mathrm{sp}} = (\sum_i e_i^2)^{1/2}$. The margin is indispensable: the gradient of $\sqrt{p_0^2 - |\mathbf{p}|^2}$ diverges near the light cone. In the source's fixed original rest chart it is preferable to retain a time-component budget throughout intermediate comparisons, as in (5), rather than assume they remain timelike.

## 5 Comparison, costs, and feasible extensions

### 5.1 What is preserved and what is added

The source already has the completed graph square, the positive $6(x-1)^2$, correctly directed boundary currents, full finite-parameter existence machinery, a conformal mass expansion, and an explicit warning about subcritical weight. We retain those mechanisms; no mathematical correction to them has been found in the inspected scope. The present additions are a signed off-equation ledger with a differentiated height residual, a boundary budget in the same variables, and explicit all-charge nonlinear tail constants on the actual conformal end. The classical Green identity and Banach theorem are credited, not renamed as new Ram theorems.

The gain over separate equation-error bounds is visible in (16): scalar curvature can be unaffected by an aligned defect even while the current error is small but nonzero. The gain over an unspecified small end error is (22): the required decay exponents, physical normal derivatives, static weight, and constants are explicit. The price is substantial: the actual state, boundary, derivative, and discarded-tail enclosures must be established. There is no measured runtime improvement, numerical solution, or automatic constraint-preserving reconstruction in this report.

The original private material [6, 7] supplies the right interpretation of local moments and inverses: finite panel coordinates are admissible only with their physical function-space norm, boundary lift, complete derivative, and a bounded omitted remainder. The present report uses those principles to define a valid certificate for the source's fields, not to replace the global geometric construction with a finite matrix.

### 5.2 Pros, limitations, and next mathematical work

The aligned ledger permits correlated residuals and one-sided margins, rather than spending an unnecessary separate absolute error on each equation. It also makes boundary mistakes visible: the curvature combination $\beta_B + \beta_w/x$ and current combination $\beta_B + \beta_w$ are different. Conversely, preserving the scalar sign does not preserve the mass budget or the selector's geometric properties; each must be checked.

The end certificate works for all four charges in a fixed chart and can retain signed finite-panel moments. Its restrictions are the exact conformal background, positive branch, a proved end expansion, and whole-end majorants. A possible extension to $g = (1+u)^4(b+k)$ would need additional metric-current terms and weighted bounds on $k, \nabla k, \nabla^2 k$; those costs are not zero and are not supplied here.

Further useful work would construct an equilibrated, boundary-compatible graph reconstruction with exact $\operatorname{div} p = 0$, or certify the derivative term in (15); prove a full weighted inverse bound for a specific geometric input; and combine the signed mass panels with verified exterior asymptotics. To extend the source's whole existence proof to approximate equations, one would additionally need uniform selector, positivity, transmission, unsafe-set, and area-transfer estimates under these perturbations. Nothing here proves those uniform statements.

### 5.3 Current primary literature and status

The assigned October 5, 2026 repository manuscript claims the global maximal AH result [1]. Captured-tree byte identity is provenance, not evidence of external acceptance or proof validity. The primary refresh on October 9 distinguishes that claim from the following accessible literature.

Chruściel–Herzlich [2] supplies the classical AH mass framework; its primary abstract and version metadata were checked, not its complete proof. Neves [3] constructs AH inverse mean curvature flow lacking convergence needed for a direct Penrose argument. Its primary abstract was checked; that obstruction does not refute a different comparison strategy. Khuri–Kopiński [4] treats constraint construction with MOTS boundary and certain perturbations of Schwarzschild–AdS. The author's primary PDF abstract and selected statements were inspected, not the full perturbation proof. Finally, Ng's September 18, 2026 preprint [5], checked at primary abstract level, states a suboptimal charged inequality and a hyperboloidal analogue. That is not the sharp maximal AH theorem with decaying $K$ and $\Lambda = -3$ used here. These access scopes do not amount to an exhaustive priority or external-proof audit.

## 6 Reproducibility and exact scope

The owned read ledger records pre-read hashes before proof displays, literal bounded intervals, unread complements, and post-read hashes for all fourteen cached public text files and two original private files. Source inclusion and heading searches are not counted as reads. Full reads cover the introduction, preliminaries, completion, paper wrapper, preamble, and README. Scientific partial reads cover the graph identities through their boundary proof; the area-transfer criterion and proof; maximal equations, mass proof, fixed-parameter existence, enclosure and conclusion; conformal approximation and selected stress/attachment passages; compact construction and transmission statements; scalar-comparison statement and corner smoothing proof. The ledger, not a claim of all-module inspection, specifies the remaining intervals. In total, 2,764 of 7,359 public text lines were literally read; the 4,595-line unread complement includes the source bibliography.

Fresh original private reads comprise the full 295-line `TR1-lin-nonlin/stability.tex`, and lines 65–197, 270–387, 492–558 of the 1066-line original TR8. The rest of TR8 and other included private modules were not read. No whole-corpus claim is made. The statement map in Section 1.2 is broader than the independently checked deduction scope.

The owned executable checks use exact rational arithmetic and explicit algebraic formulas. They test actual one-dimensional graph metrics with nonzero height residual, boundary normal factors, aligned defect budgets, static radial flux and tail cancellation, nonlinear divided-difference majorants, and parameterized mass/area intervals. The final owned run passed 1,122 exact-rational cases and 3,389 assertions, with no numerical-tolerance tests. Counts, code hashes, and limitations are in `_audit/M516-tests.json`. These finite fixtures do not verify global Jang existence, stress saturation, weak flow, or the Penrose theorem. Parent checks are excluded from the author counts.

The report preserves applicability L3. The independent contribution assessment is C2 because the useful refinements are local consequences of already sophisticated source identities with classical norm and Green-function tools. No new sharp geometric inequality, global continuation theorem, or effective construction is claimed.

## References


[1] OpenAI mathematics repository, *The Penrose inequality for maximal asymptotically hyperbolic initial data*, October 5, 2026. Assigned cached manuscript, fourteen text files; fixed captured source identified in the owned ledger. `https://github.com/openai/math`.

[2] P. T. Chruściel and M. Herzlich, *The mass of asymptotically hyperbolic Riemannian manifolds*, Pacific J. Math. 212 (2003), 231–264. `https://arxiv.org/abs/math/0110035`.

[3] A. Neves, *Insufficient convergence of inverse mean curvature flow on asymptotically hyperbolic manifolds*, J. Differential Geom. 84 (2010), 191–229. `https://arxiv.org/abs/0711.4335`.

[4] M. Khuri and J. Kopiński, *Asymptotically hyperbolic Einstein constraint equations with apparent horizon boundary and the Penrose inequality for perturbations of Schwarzschild–AdS*. `https://doi.org/10.1088/1361-6382/acb24b`. Primary author PDF inspected at the stated limited scope.

[5] E. Ng, *The suboptimal Penrose inequality for charged initial data sets*, arXiv:2609.22418, September 18, 2026. `https://arxiv.org/abs/2609.22418`.

[6] M. Subbarao, original private `RamNonlin-IE-PDE-astra/TR1-lin-nonlin/stability.tex`, 295-line local original, freshly read in full. Bounded-isomorphism ball, residual, boundary, and constraint proofs. Confidential source; no online transmission.

[7] M. Subbarao, original private `RamNonlin-IE-PDE-astra/TR8-lin-non-lin/TR8-lin-non-lin.tex`. Fresh scoped reads: 65–197, 270–387, 492–558. Complete PDE domain, nonlinear inverse, and physical variational residual proofs. Confidential source; no online transmission.